

3.1

Algebra 1 · Module 3

relation

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domain

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range

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independent variable

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dependent variable

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mapping diagram

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function

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Algebra 1 · Module 3

function notation

domain

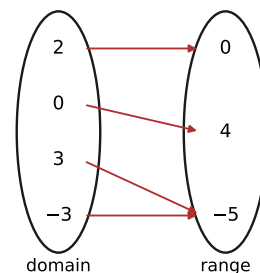
The set of the first numbers of the ordered pairs — the x-values (inputs).

Domain above: $\{-3, 0, 2, 3\}$

relation

A set of ordered pairs.

$\{(2, 0), (0, 4), (3, -5), (-3, -5)\}$



independent variable

The input variable — its value is chosen. Usually x, on the horizontal axis.

Time, in "texts sent over time"

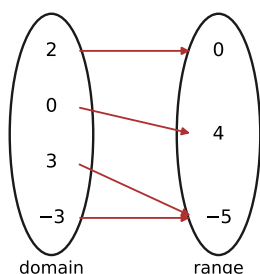
range

The set of the second numbers of the ordered pairs — the y-values (outputs).

Range above: $\{-5, 0, 4\}$

mapping diagram

A picture of a relation: domain in one oval, range in the other, with arrows pairing each input to its output.



dependent variable

The output variable — its value **DEPENDS** on the independent variable. Usually y, on the vertical axis.

Texts sent, in "texts sent over time"

function notation

A way of writing an equation so that $y = f(x)$, pronounced "f of x."

$$f(x) = -2x + 9 \rightarrow f(4) = 1$$

function

A relation in which each element of the domain is paired with **EXACTLY ONE** element of the range.

An x-value never repeats with a different y.

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vertical line test

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discrete function

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continuous function

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linear function

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nonlinear function

3.4

Algebra 1 · Module 3

x-intercept

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Algebra 1 · Module 3

y-intercept

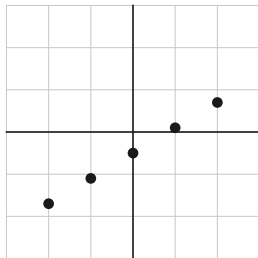
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positive (function)

discrete function

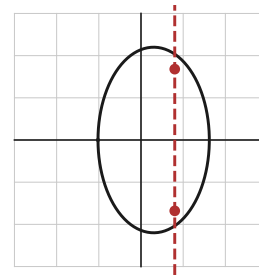
A function in which the points on the graph are NOT connected.



Number of books bought: 1, 2, 3, ...

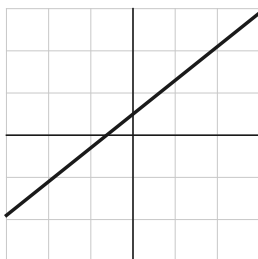
vertical line test

If any vertical line crosses the graph more than once, the relation is NOT a function.



linear function

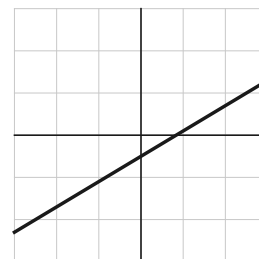
A function whose graph forms a line.



$$f(x) = 0.8x + 0.5$$

continuous function

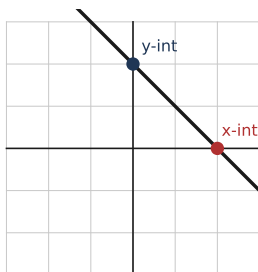
A function that can be graphed with a line or a smooth curve — no breaks.



Distance over time

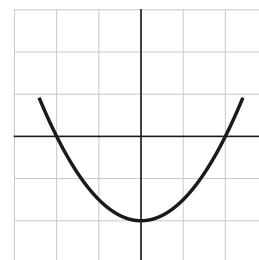
x-intercept

The point at which the graph crosses the x-axis. There, $y = 0$.



nonlinear function

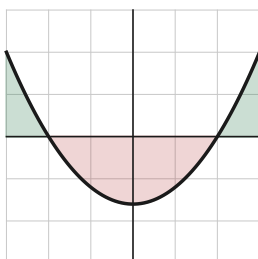
A function whose points cannot all lie on one line.



$$f(x) = 0.5x^2 - 2$$

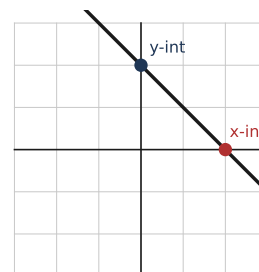
positive (function)

Where the graph of a function lies ABOVE the x-axis ($f(x) > 0$).



y-intercept

The point at which the graph crosses the y-axis. There, $x = 0$.



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negative (function)

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root

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zero

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line symmetry

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increasing

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decreasing

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extrema

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relative maximum

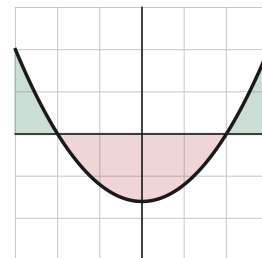
root

A solution of an equation.

Roots of $x^2 - 4 = 0$: -2 and 2

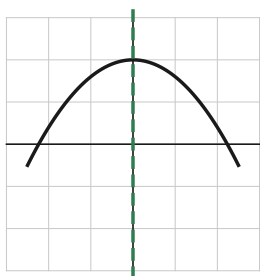
negative (function)

Where the graph of a function lies BELOW the x-axis ($f(x) < 0$).



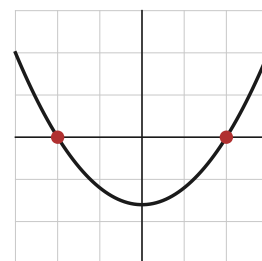
line symmetry

A graph has line symmetry if, folded along a line, one side matches the other exactly.



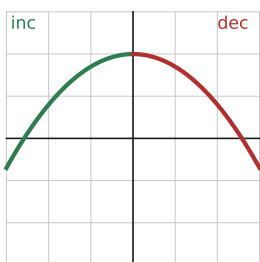
zero

An x-intercept of the graph — the value of x for which $f(x) = 0$.



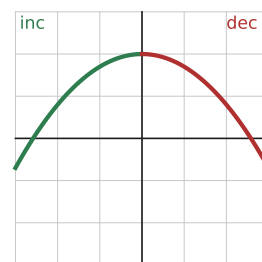
decreasing

Where the graph of a function goes DOWN as you read from left to right.



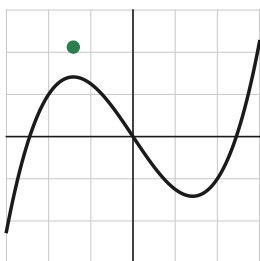
increasing

Where the graph of a function goes UP as you read from left to right.



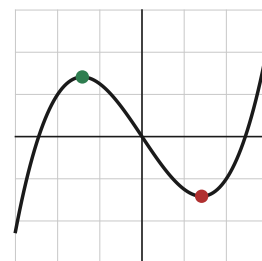
relative maximum

A point higher than every point near it — the top of a hill.



extrema

Points that have the greatest or least function values — relative maximums and minimums.



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relative minimum

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end behavior

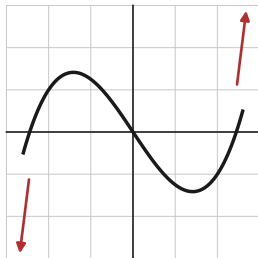
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**key features
(sketching)**

end behavior

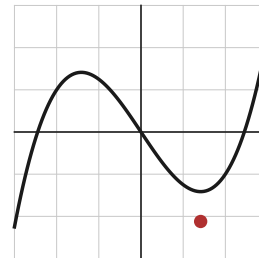
What the graph does at the far left and far right of its domain — as $x \rightarrow -\infty$ and $x \rightarrow \infty$.



Here: down on the left, up on the right.

relative minimum

A point lower than every point near it — the bottom of a valley.



key features (sketching)

To sketch from a story: intercepts · linear or nonlinear · positive / negative · increasing / decreasing · extrema · end behavior.

Read the context first — it decides each feature.